Personal Reflection through Interactive Data Comics: A Story on Reading Data

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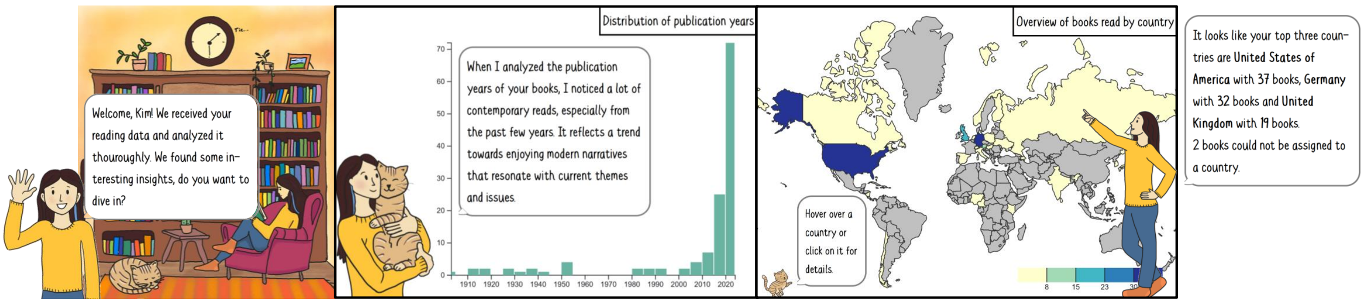

Figure 1: Excerpts of the interactive data comic on personal reading data: introduction and setting (left), books by publication years (center), books by location (right); collage of cut-outs, see Figure 2 for the entire comic.

Abstract

Data comics are an appealing medium for visualizing personal taste and interests. Such comic-style, data-driven stories can further be extended regarding interactive data exploration, adaptive storytelling, and an overall personalized experience. This paper presents an interactive data comic that analyzes personal reading data. It adapts both content and structure to users' data and selections. Combining elements of data visualization, sequential storytelling, and user interactivity, the comic transforms the data into a relatable narrative organized in different chapters. Accompanying the visualizations, data analysis guidance and book recommendations are generated specific to the personal data through a large language model. An explorative user study with six participants evaluated the impact of personalization and interactivity on user experience. Results show the accessibility and appeal of the developed comic, as well as that the narrative format fosters self-reflection.

Index Terms: Personal visualization, personalization, data-driven storytelling, data comics

1 Introduction

From daily habits to fitness and health tracking, personal data holds valuable insights. Current approaches are often related to *optimizing* (e.g., steps taken, sleep quality indices) and are presented through aggregate visualizations in dashboards, being displayed on mobile devices for regular checking. However, this approach is not suitable for all types of personal data, particularly those that lack crisp objectives. Examples of such data include all information related to tastes and moods, interests and preferences, as well as habits and routines. We consider book reading data—*when did I read which book?*—an exemplar for this kind of personal data and explore it in this work. Here, a rather contemplative approach to data analysis is more suitable, not a monitoring tool but an interface to show different aspects and examples, inspiring new reading directions or recognizing personal biases.

Matching this scenario, data-driven storytelling and narrative visualizations [14, 8] provide a structured way to communicate data and insights. Such data stories are typically designed for broad audiences, which matches our goal that everybody could use them without instruction. However, most data stories are generic and relatively static, rather than tailored to individual users or supporting interactive exploration. In line with recent research to make data stories more interactive [19, 9] and personal [7, 5, 9], we explore *data comics* [1, 2, 19] as a specific medium for presenting personal reading data. Such comics combine data visualization, sequential storytelling, and user interactivity with engaging, playful comic elements and characters. This medium, we believe, could support the contemplative nature of our scenario, especially if framed accordingly in a fitting analysis setting and providing options to interact with and explore the data.

Our paper presents a web-based data comic that visualizes users' personal reading data. It integrates various interaction approaches, enabling adaptive storytelling based on user interests and selections, thereby contributing further to a personalized reflection of reading behavior. As shown in Figure 1 and Figure 2, the story starts by introducing the setting and characters—*Rose*, the main character, and her cat *Archie*. Then, the user can select from four different chapters (indicated by a differently colored background), each including visualizations along with generated comments on the data and interactive options to explore deeper. The aspects covered include book genres, temporal and geographic distributions, as well as language and author gender. The chapters can be read in any order, or skipped. Finally, the comic concludes and gives the option to further converse with *Rose* about the data (AI-based chat client). A user study was conducted to assess user experience and value of the personal data reflection. Six participants, working with their own reading data, provided rich feedback; generally, they gained insights of personal relevance and found the presentation format enjoyable and appealing. The interactive comic is publicly available with sample data at `https://personal-datacomic.web.app/`.

2 Related Work

Interactive storytelling can adapt to user preferences by tailoring story elements or generating new storylines [16]. This encourages engagement and exploratory reading, and allows users to actively participate in the narrative. It also increases motivation by catering

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Figure 2: Entire data comic (left) with annotated sections (right); state after all story parts were read (without outro).

to personal styles and emotional qualities. Furthermore, it allows for reusability of materials, enabling the creation of different stories and games from a common set of story units, patterns, and structures [11]. Many interactive storytelling approaches rely on branching storylines [11, 16]. It was found that adapting the course of the story to user characteristics can effectively improve user satisfaction and experience [16].

Bach et al. [1] introduced data comics as a storytelling format that integrates data visualization, sequential flow, and narrative elements into a comic-style panel structure with characters, callout bubbles, captions, etc. Wang et al. [19] expanded this concept by defining six interactivity goals in data comics, including *navigation*, *details on demand*, *changing perspective*, *branching paths*, *pause and reveal*, and *input of data*. It was demonstrated that these goals can help to achieve personalized data storytelling, as the example *Personal CO2 Emissions* [19] shows. Gomez Ortega et al. [7] presented a project where participants were encouraged to create data comics from their personal data sets. In this case, participants were requested to "contest the data" or "chat with the data", drawing relations between the data, connecting data to their lived experiences, and correcting or opposing where necessary.

Personal data analysis approaches are mostly about *visualizing* and—in many cases—*optimizing* personal behavior. Typical areas of application are sports (e.g., [20]) and health (e.g., [10]), but also examples from other areas such as web browsing [4] or sleeping [12]. A less commonly discussed use of personalization is the reflection of personal experiences or impressions. These approaches do not focus on optimization, but rather only on the presentation of personal data, such as memories of places visited [18]. A related approach that deals with media consumption, as we do in this paper, is *MoViQuotes* [13], where users can visualize their personal impressions of a movie abstractly using colors and quotes, which can then be shared. Reflecting on personal experiences has also become popular on media platforms. A well-known example is *Spotify Wrapped*, a simple annual review that evaluates the listening statistics of a Spotify user and contextualizes the data playfully [17]. For reading, *GoodReads* offers *Year in Books*, an annual review of reading behavior with summarized statistics [6]. *Bookly* tracks reading behavior and creates a shareable report for each book [3]. Most similar to our approach, *Bookstats* generates detailed statistics [15], but not in comic or storytelling format and not as exploratively.

3 Personalization Approach

In this paper, we combine such personal summaries with the idea of data-driven storytelling through data comics. We propose two different methods to achieve a personalized data story: First and foremost, the story shows and adapts to the user's *personal data*, the subject of the story. It is represented by visualizations, but also reflected by adaptively generated textual statements and used for personal book recommendations. But second, the user can further customize the story to their *preferences* using selections. To support both personalization aspects, the narrative structure was designed to be modular and adaptable. Rather than following a fixed sequence, the story is composed of sections that can be arranged, emphasized, or even omitted. This flexible structure allows for different users to receive different versions of the story. It enables both system-driven and user-driven personalization—the system automatically adjusts the contents or the users choose themselves. To facilitate and structure the planning and subsequent implementation of the personalized data comic, we use a graph-based model. Figure 3 depicts this story graph for the developed data comic, showing where the story can take different paths and whether the decision about the path is made by the user or by the system.

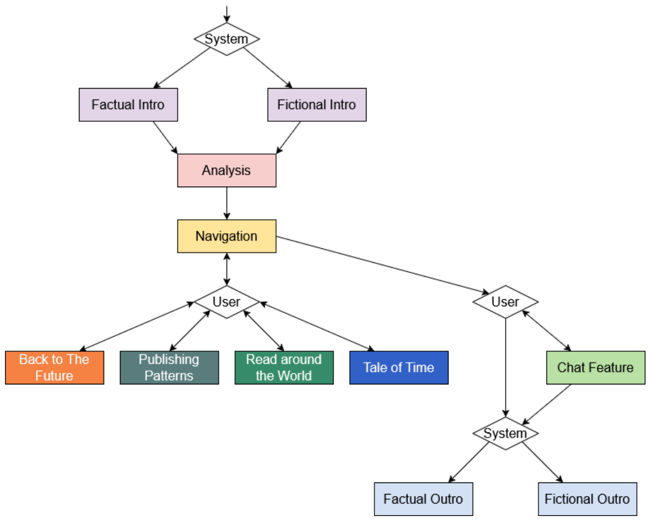

Figure 3: Story graph of the personal data comic, with content parts as rectangles and decisions by the system or user as diamonds.

4 Personalized Interactive Data Comic

The intent of the designed data comic is to encourage self-reflection through personal data, in particular, discovering one's own reading preferences and potential biases. It aims to encourage users to look at data playfully, making the data accessible and relatable. A female character (*Rose*) acts as a friend to the reader, discussing their reading data with them. A cat (*Archie*) is introduced as a sidekick, who lightens the moods and helps the reader in using the interactive comic. The data comic has a lighthearted tone, with a character-driven narrative to create a personal atmosphere.

It is implemented as a *React* web application, using *d3.js* for interactive visualization and the API of *OpenAI* for requests to a large language model. While, for adaptivity, text generation for the comic is supported through generative artificial intelligence, all artistic elements, including the comic characters, were hand-drawn by the main author of this work.

Personal Reading Data As input to the data-driven story, we rely on popular reading platforms. Services such as *Goodreads* and *StoryGraph* offer readers various features to track reading. Users can manage the books they are reading, have read, and want to read. They can rate and review books, get recommendations, and check their reading statistics. The platforms offer the user to export their data in CSV format, which includes a list of books with their current reading status (*to-read*, *currently reading*, *read*), completion date, rating, review (if available), title, author, ISBN, as well as potentially additional metadata such as the number of pages, the average rating, the publication year, and more. Specifically, we used the reading data from *Goodreads* or *StoryGraph* (piped through *Goodreads* for providing richer metadata) as the foundation and enriched it in a semi-automatic approach with additional publicly available information from *Wikidata* and *Wikipedia* such as genre, subject, as well as author country of origin and gender.

Adaptive Story As shown in Figure 4, the comic has a modular narrative structure that allows different readers to be presented with different outcomes based on their personal data. At the story opening, there are two different options. The *Factual Intro* explains the data comic as a genre and its goals. In contrast, the *Fictional Intro*, as shown in Figure 2, sets the story in Rose's house on a rainy afternoon. In both cases, the story continues with an initial *Analysis* of the user's data, followed by a series of chapters, each dealing with a different aspect. After reading any number of chapters in any order (*Navigation*), users can explore an optional *Chat Feature*, where they can ask the character questions about their reading data. In this way, they can extend the reflection of the reading data together with the (artificial) character. At the end of the story, again, there are two options: The *Factual Outro* concludes by saying goodbye. The *Fictional Outro* picks up the rainy day motif and bids the user farewell against a backdrop of improving weather, symbolized by a rainbow. While the decision for following the fictional or factual intro and outro is taken by the system (based on an analysis of the provided book list, see below), the chapter navigation and chat is interactively controlled by the users.

Data-driven Contents Structuring the data-driven analysis, the main chapters offer complementing perspectives on the reading data, supporting different reflection perspectives. The users choose which chapters to read, and in what order, through a navigation that resembles a display of different books (*Navigation*; see annotations to Figure 2). The chapters of the comic are colored differently to make them easier to distinguish. The chapter *Back to the Future* focuses on the list of books the user wants to read in the future and presents it as a stack (stacked bar chart) where users can retrieve details on each of the books; the chapter offers personalized book recommendations and advice on how to tackle the list. The chapter *Publishing Patterns* reveals hidden trends in the books that were read, such as author gender ratios (donut chart), genre preferences (word cloud), and publication year distribution (temporal bar chart). The chapter *Read around the World* provides an overview of the countries of origin of the books read (choropleth world map) and the language distribution (pie chart). The chapter *Tale of Time* discusses an interactive timeline of the user's reading history, first per year (bar chart), but also per month on selecting a year (stacked temporal bar chart). While each chapter has an individual structure, they all center around a certain aspect of the data, with *Rose* commenting on the data and the cat *Archie* providing useful hints on exploration options.

Interactive Visualization As mentioned above, the data comic includes various visualizations of the personal data, at least one per chapter, but often multiple. They rely on simple, easy-to-understand encodings, such as bar charts, pie and donut charts, word clouds, or choropleth maps. The visualizations usually offer base interactions for retrieving details on demand by hovering over the respective visual element. Beyond that, in some visualizations, elements can be selected, with further information or a new visualization showing in the next panel. To make the reader aware, *Archie* hints at these interaction options. For example, in the word cloud visualization and the map (which encode the number of read books per genre or country, respectively, in *Publishing Patterns* and *Read around the World*), users can select a genre or country, then see a *stack of books* (stacked bar chart); in turn, clicking on a book (rectangle in the stacked bar chart) reveals the detailed metadata of the book in another panel. In that way, and with the characters also embedded in the visualization panels, the visualization and comic elements blend into a coherent representation.

Text Generation and Recommendations Across all chapters, the comic supports reflection by providing commentary on the data through the character of *Rose*. For instance, as illustrated in Figure 1, the comments summarize temporal (*Publishing Patterns*) or geographic (*Read around the World*) patterns, and accompany the respective visualization of the data, or support the *Analysis* panel at the start (see Figure 2). These adaptive elements are generated through a large language model (*GPT-4o-mini*, as a fast and cost-efficient model). To avoid waiting times while reading the comic, they are prepared on loading the data initially. Recommendations for new books to read can also be requested interactively but are computed on the fly using the same language model (e.g., considering a selected country in *Read around the World*). For recom-

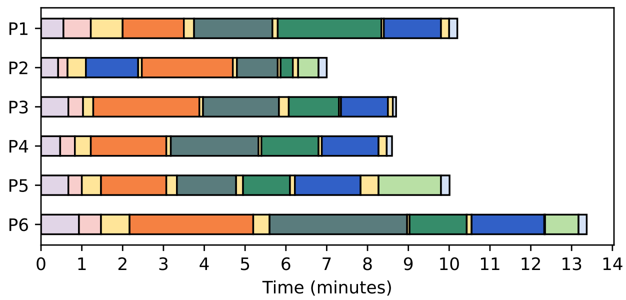

Figure 4: Visualization of user sessions (**P1**–**P6**), sequence and usage duration of sections of the comic: *Factual Intro*/*Fictional Intro*, *Analysis*, *Navigation*, *Back to the Future*, *Publishing Patterns*, *Read around the World*, *Tale of Time*, *Chat Feature*, *Factual Outro*/*Fictional Outro*.

mendations, we provide the previously read books, and rely on the trained knowledge of the model for suggesting fitting recommendations. For the *Chat Feature*, to further talk with *Rose* about the data, a text input allows the user to ask any question, while *Rose*'s AI-based replies are presented in a comic panel, creating the impression of a living character. In all requests to the language model, through prompt engineering, we tried to ensure character consistency (defining *Rose* as *friendly bookworm*) and brevity in replies to fit the limited space in the comic; suggesting specific opening phrases for specific panels further helped to control the consistency of the generated text.

5 User Study

With the goal to understand the usage of the personalized data comic and to assess the user experience as well as the support to reflect on the data, we conducted an explorative user study with six participants (**P1**–**P6**; recruited from the personal environment of the main author; all female). Each participant examined their personal data comic based on their individually provided reading data. Results were collected through recording of user behavior and semi-structured interviews. First, the participants were encouraged to read the personal data comic as they would do on their own, following their preferences and interests, while commenting on their approach (*thinking aloud*). Second, in the interview, their overall impressions were recorded, including user's opinion on the usability, positive and negative aspects of the prototype, and commentary on performed interactions. The study lasted about 45 minutes and also included more general questions on personalization—for brevity, we focus here on the results directly related to the presented data comic.

Usage Figure 4 shows that participants took different times to read the comic (approximately 7–13 minutes). The reading time is relatively evenly distributed across the four chapters, with no chapter clearly dominating. We observe some variation between participants (e.g., **P1** spending longer time in *Read around the World*, **P2** less), but overall no stark contrasts. Although the participants were free to choose any sequence of chapters, almost all followed the suggested reading sequence (except for **P2**, who read *Tale of Time* first). In contrast, the *Chat Feature* was used by only half of the participants (**P2**, **P5**, **P6**). Overall, these results suggest a balanced chapter structure. The consistent chapter sequence might be an artifact of the artificial study setting (as hinted at by some participants, who felt observed and did not explore as freely). Nevertheless, while reading, the participants actively reflected on their reading behavior. They connected the data with memories. For example, one participant noticed a spike in the reading timeline, relating it to a vacation. When participants noticed misalignments with their recall, they discussed that. New insights were encountered with expressions of surprise and interest. The chat feature was—as intended—rather considered as optional, but also the participants who tried it remarked that they did not have specific follow-up questions.

User Experience All participants fully agreed with statements that the comic is *easy to use* and *appealing* (both, average of 5 on a scale from 1 – *lowest* to 5 – *highest*). Somewhat mixed, but still overall positive were agreements to respective statements on *usefulness* (avg. 3.25), *efficiency* (avg. 3.25), and *engagement* (avg. 3.5). Lower scores for usefulness and efficiency are not surprising as the contemplative setting of the comic—unlike for other visualization tools—does not suggest clear goals of analysis, as also confirmed by user feedback from the interviews; the participants rather characterized the approach as *entertaining*. Considering engagement, participants remarked that the story remains in the background and is not that exciting. There is also little connection between the individual chapters, but participants interested in the comic's data insights still found it engaging.

Further Feedback Generally positive feedback was given on the design of the data comic and the characters. Furthermore, participants appreciated the interactive and structured format of the comic. For example, the division into chapters, which were revealed step by step, and the ability to explore details were mentioned. The most frequent criticism was that the comic contains too much text in some panels. Moreover, some participants missed the comparison to other readers. A controversial feature was the AI-generated book recommendations; some participants did not find the personal recommendations fitting, while others liked theirs.

Limitations The study is limited by a small sample size and a non-comparative design. We prioritized ecological validity and an explorative approach over quantifiable insights. Recruitment was challenging, as we included only participants who had available personal reading data on one of the two supported platforms and were willing to share it. Despite the efforts to create a realistic scenario, some still perceived the study setting as artificial.

6 Conclusion

This paper introduced a data comic that adapts to users' personal reading data, combining storytelling, interactivity, and further personalization aspects. It demonstrates that the genre of data comic well suits the rather contemplative use case of analyzing personal reading data. Future work could expand personalization options, explore other domains of personal data, and include larger and more diverse user studies. Further integration of generative AI could allow for even more adaptive and engaging storytelling, dynamically shaped by users' data and preferences. Recent progress in image generation and style transfer might even allow the dynamic generation of consistent character artwork.

Overall, personalized interactive data comics are a promising direction for presenting personal data in appealing and contemplative ways, through the lens of a narrative. While this paper presents only one such example, further examples can be developed following a similar approach and learning from our experience. For instance, we suggest splitting the data into relevant facets or perspectives that form the user-selectable chapters of the comic. Visual representations and interactions should be kept simple and understandable, but the comic itself can provide hints on how to read and use the visual data representations. Relying on a small set of characters with clearly assigned specific roles, the data analysis can be embedded into a story that sets a frame for the reflection, indicating that the goal is not to optimize but to think about the data.

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